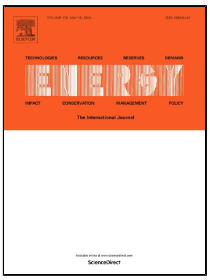
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Improved Performance of Ferroelectric Nanocomposite Flexible Film Based Triboelectric Nanogenerator by Controlling Surface Morphology, Polarizability, and Hydrophobicity

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Abstract

In recent years, triboelectric nanogenerator (TENG) has been attracting lots of attention for harvesting electrical energy from mechanical energy, because of its simplicity in designing, cost-effectiveness, and high output power. In the present trend, the performance of TENG is improved mainly by surface modification using complex techniques. In the present work, we have demonstrated enhanced triboelectrification between polyvinylidene fluoride (PVDF) and polytetrafluoroethylene (PTFE) by incorporating ZnO nanorods into PVDF polymer. The fabricated ZnO-PVDF/PTFE based TENG showed 21% enhanced output voltage and 60% enhanced short-circuit current in comparison to PVDF/PTFE based TENG with an instantaneous output power density of ~10.6 μW/cm^2. The increase in triboelectrification comes not only from the enhancement in β-phase content, which leads to increase polarizability, but also from the enhancement in surface roughness, hydrophobicity, and a decrease in work function of PVDF after incorporating ZnO nanorods.

Keywords: Nanogenerator, Triboelectric, Ferroelectric, Piezoelectric, PVDF

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1. Introduction

Triboelectric nanogenerator based on triboelectrification is the new emerging clean and effective method to generate electrical energy from to be wasted mechanical energy. The triboelectric effect is known to the scientific community for a long time. However, the charge induced from this effect was considered to be a negative effect by the scientific community and the engineers [1–4]. Recently with the realization of the potentiality of using this effect for harvesting electrical energy from mechanical energy, this physical phenomenon has been attracting lots of attention. The triboelectric nanogenerator offers an effective method of energy harvesting having high energy density output with a very simple design [5–7]. The mechanism and the factors influencing the working of triboelectric nanogenerator have not been yet well understood. In order to scale up and enhance the output power of the triboelectric nanogenerator, understanding the factors influencing the output of the triboelectric nanogenerator is very important. Currently, researchers have tried and improved the efficiency of the triboelectric nanogenerators either (1) by choosing two different materials having a large difference in triboelectric effect, (2) by increasing the contact surface area, or (3) by increasing the surface roughness [8,9]. The surface roughness of the triboelectric materials has been enhanced by argon plasma treatment [10,11], micro-roughness with nanowires [12], inkjet-printing [13], transferable printing method [14,15], electrospun method [16,17], and nanoparticle dispersion [18]. The above processes are time-consuming, and some of the processes mentioned above require the application of high voltage. Therefore, there is a need for simple, effective and less energy consumed approach for enhancing the triboelectric property of the material.

Recently Polyvinylidene fluoride (PVDF) got attention to use as one of the active material for the triboelectric nanogenerator because of its high flexibility, easy to process and easy to change its properties by different film preparation techniques or by incorporating

nanostructures into it [19–22]. PVDF is an electroactive semicrystalline organic polymer having a high piezoelectric coefficient as compared to other polymers [23–26]. PVDF exhibits five different complex structures according to their molecular conformations [27–29]. Out of the different phases, β-phase has the highest electric dipole moment [30]. Therefore, β-phase is of great interest for PVDF to be used for sensors, actuators, energy harvesting applications. K. Xia et al. [21] have even shown a temperature sensor based on the PVDF based triboelectric nanogenerator. Recently many attempts have been made to change the property of PVDF for using in triboelectric nanogenerator such as by incorporating MoS_2 [22], Fe_3O_4 nanoparticle [20], $ZnSnO_3$ [31] or film synthesis by electrospinning method [32]. However, these methods have been employed just to increase the β-phase content of PVDF which in turn leads to an increase in polarizability/dielectric constant of PVDF. It has not been attempted to change the other properties of PVDF like surface roughness, hydrophobicity and work function that can also greatly influence the output of triboelectric nanogenerator. ZnO is, on the other hand, a biocompatible material with various unique and tunable properties including piezoelectricity, possesses the highest number of different morphologies with easy to synthesize and cheaper [33,34]. In the present work, we have demonstrated modification of the surface topography, polarizability, and hydrophobicity of PVDF by incorporating ZnO nanorods with the aim to use it for efficient triboelectric nanogenerator. The novelty of our work is that we have enhanced the triboelectric effect of PVDF, only by introducing ZnO into the polymer matrix without any further surface treatment or any complex synthesis technique. We have fabricated two different (a) PVDF/PTFE and (b) ZnO-PVDF/PTFE based sliding TENG using wooden support and tested their performance by measuring open circuit voltage, short circuit current, and instantaneous output power at different load resistances. The enhanced triboelectric effect is correlated with the increase in surface roughness, polarizability, and hydrophobicity of PVDF with the incorporation of ZnO in PVDF.

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2. Experimental

2.1 Preparation of ZnO nanorods: ZnO nanorods were prepared using the hydrothermal method. For the synthesis, 1.6 g of zinc sulfate ($ZnSO_4$) was dissolved in deionized water (DI) (25 ml). Another precursor solution was prepared by dissolving sodium hydroxide (NaOH) (4 g) in DI water. The two solutions were mixed together. When we added the sodium hydroxide solution, the color of the solution changed into a milky color. However, after stirring for 3 min, we got a clear solution, and after that only we added another drop of sodium hydroxide solution. The process continued until we added all the solutions of sodium hydroxide. The prepared solution was then transferred to a 100 ml Teflon beaker, and it was sealed inside a stainless steel autoclave which was kept in an oven for 12 h with the temperature maintained at 120 °C. The solution was cooled down and filtered to obtain the prepared ZnO nanorods.

2.2 Preparation of the nanocomposite films of ZnO-PVDF: 1 g of PVDF powder obtained from Alfa Aesar (Product Number 44080) was dissolved in N,N-Dimethylformamide (10 ml). A 15 wt.% of ZnO, thus prepared above were mixed thoroughly first by magnetic stirring, followed by ultrasonicating the solution for 30 min, thereby enabling ZnO nanorods to disperse in the solution thoroughly. The solution was then drop casted over a clean glass substrate, and it was dried in an oven which was maintained at 90 °C for 1 h. The film was later removed from the glass substrate to get the flexible and freestanding PVDF and ZnO-PVDF nanocomposite films. The size of the films was 2.5 cm × 3 cm with the thickness of the films as ~30 μm.

2.3 Characterization. The structural analysis of ZnO, PVDF, and their nanocomposite films were performed using a Rigaku Ultima –IV X-ray diffractometer (XRD). For analyzing the topographical structure of ZnO-PVDF, we used a JEOL JEM-2200FS transmission electron

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microscope (TEM). The vibrational characteristics of the PVDF and its nanocomposite films were characterized using a NICOLET 5700 Fourier transform infrared spectroscopy (FTIR). The surface topography and charge density of the films were analyzed using a Bruker Dimension Icon scanning probe microscope in tapping AFM mode, piezoelectric force microscopy (PFM), and Kelvin probe force microscopy (KPFM) mode respectively. The scan rate and lift height that was fixed during the measurement was 0.802 Hz and 60 nm for both the samples to obtain the surface potential image. The contact angles were measured using a Krüss DSA255 drop shape analyzer. Keithley 6514 electrometer was used for the measurement of the output voltage while for measuring the short circuit current we used a Keithley 6487 picoammeter.

3. Results and discussion

The crystal structure and phase of the ZnO and the PVDF composite films were analyzed using XRD spectra. Fig. 1 shows the XRD pattern of ZnO, pristine PVDF, and ZnO-PVDF nanocomposite film. ZnO exhibits a hexagonal phase (JCPDS card number 36-1451). The peaks corresponding to the different phases (α and β-phases) of PVDF composite film have also been identified [23,35]. The XRD pattern clearly shows that the intensity of the α-phase peaks is more in PVDF film while it has less intensity in the case of ZnO-PVDF nanocomposite film which indicates that there is a phase inversion from α-phase to β-phase after the ZnO incorporation. Therefore, from the XRD analysis, we can conclude that the electroactive phase, i.e. β-phase of PVDF has been enhanced. However, to find the relative increase in β-phase content, it is required to perform Fourier transform infrared spectroscopy (FTIR) measurement. While the inset of the figure shows the TEM images of ZnO-PVDF composite film. For observing the TEM images of ZnO-PVDF nanocomposite, we drop casted the ZnO-PVDF solution on the copper grid. The ZnO nanorods can be seen embedded into the PVDF matrix, forming nanocomposite of ZnO-PVDF.

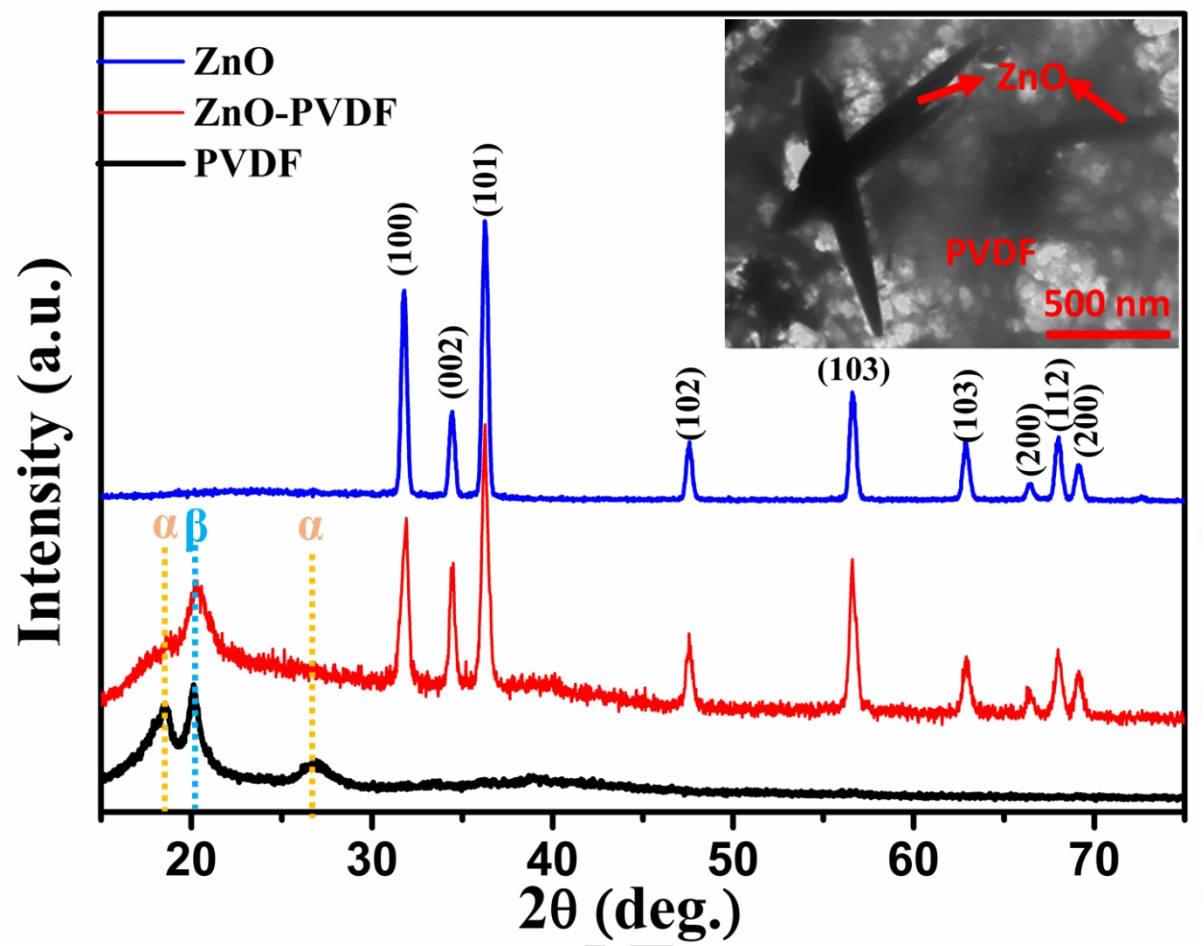


Fig. 1 XRD patterns of hydrothermally synthesized ZnO nanorods, pristine PVDF and ZnO-PVDF nanocomposite film. The TEM image of the ZnO-PVDF nanocomposite film is shown as an inset in the figure.

In order to quantify and confirm the enhancement in the β-phase content, Fourier transforms infrared (FTIR) spectroscopy measurement was done. The FTIR spectrum of PVDF and its nanocomposite film is shown in Fig. 2(a). The peak at 762 and 840 cm^{-1} corresponds to the signature peak of α and β-phases, respectively. The other peaks at 795, 974, 1149, 1211 and 1382 cm^{-1} represent the α-phase. The amount of β-phase presents can be obtained using Beer-Lambert law, $F(\beta) = \frac{A_\beta}{(k_\beta/k_\alpha)A_\alpha + A_\beta}$, where k_β and k_α are the absorbance coefficient at 840 and 762 cm^{-1}, and $k_\alpha = 7.7\times10^4$ cm^2/mol and $k_\beta = 6.1\times10^4$ cm^2/mol, respectively [24,36,37].

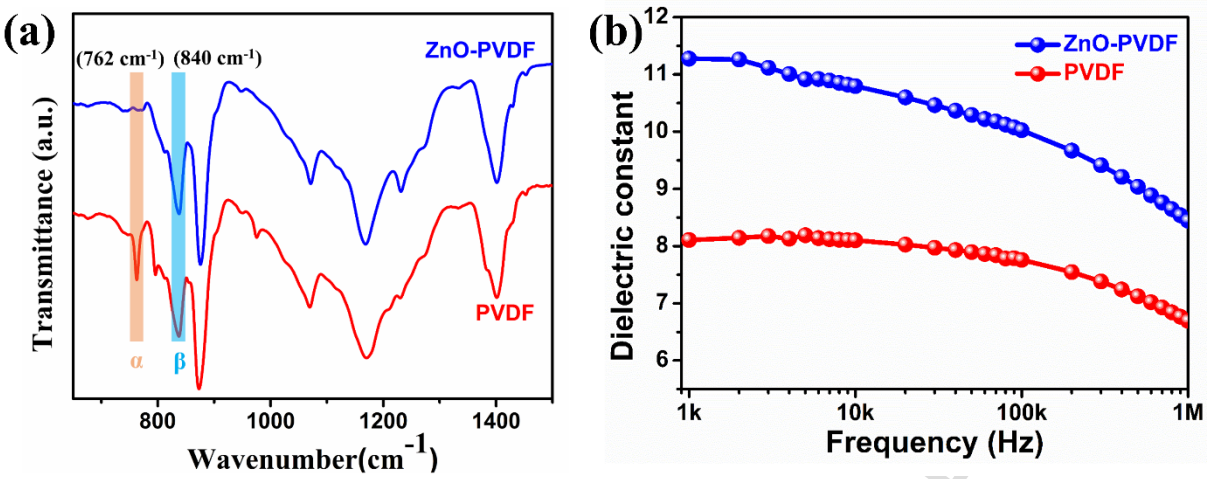


Fig. 2 (a) Fourier transform infrared spectrum of PVDF and ZnO-PVDF nanocomposite film and (b) variation of dielectric constants of the films with respect to change in frequency from 1 kHz to 1 MHz.

The value of *F(β)* for PVDF and ZnO-PVDF film was found to be 48.23% and 76.14%. This enhancement of *F(β)* in PVDF with ZnO incorporation is because of the interfacial interaction of the surface charges of ZnO with the dipoles of PVDF [38,39] which leads to the change of dominantly present alternate *trans* and *gauche* molecular conformation, i.e. *TGTGTG* (α-phase) to *all-trans* molecular conformation, i.e. *TTTTTT* (β-phase) [40–43]. As the β-phase content is increased in ZnO-PVDF nanocomposite film, it is expected that the electric dipole moment of the ZnO-PVDF will be increased as compared to PVDF film. In another word the polarizability of the PVDF will be increased. According to the Clausius-Mossotti relation, the dielectric property of the material correlates to the polarizability of the materials [44]. Therefore, as from XRD and FTIR analysis, the β-phase content is increased which directly means that polarizability of PVDF is increased, the dielectric constant of PVDF will also be increased. In order to check the change in the dielectric properties of PVDF after introducing ZnO, we measured the dielectric constant of PVDF and ZnO-PVDF nanocomposite films. As shown in Fig. 2(b), the dielectric constant is more for the ZnO-PVDF nanocomposite film. The value of dielectric constant of PVDF and ZnO-PVDF nanocomposite film was found to be 8.1 and 11.3 at 1 kHz. The value of the dielectric constant decreases with

the increase in frequency because at higher frequencies the dipole moments cannot follow the externally applied oscillating field. The nonsymmetrical structure of β-phase of PVDF is responsible for its increase in polarizability, it also can be viewed as the increase in the ferroelectric/piezoelectric property of PVDF. In order to confirm the enhancement in the ferroelectric/piezoelectric property of PVDF, we performed piezo force microscopy (PFM) measurement.

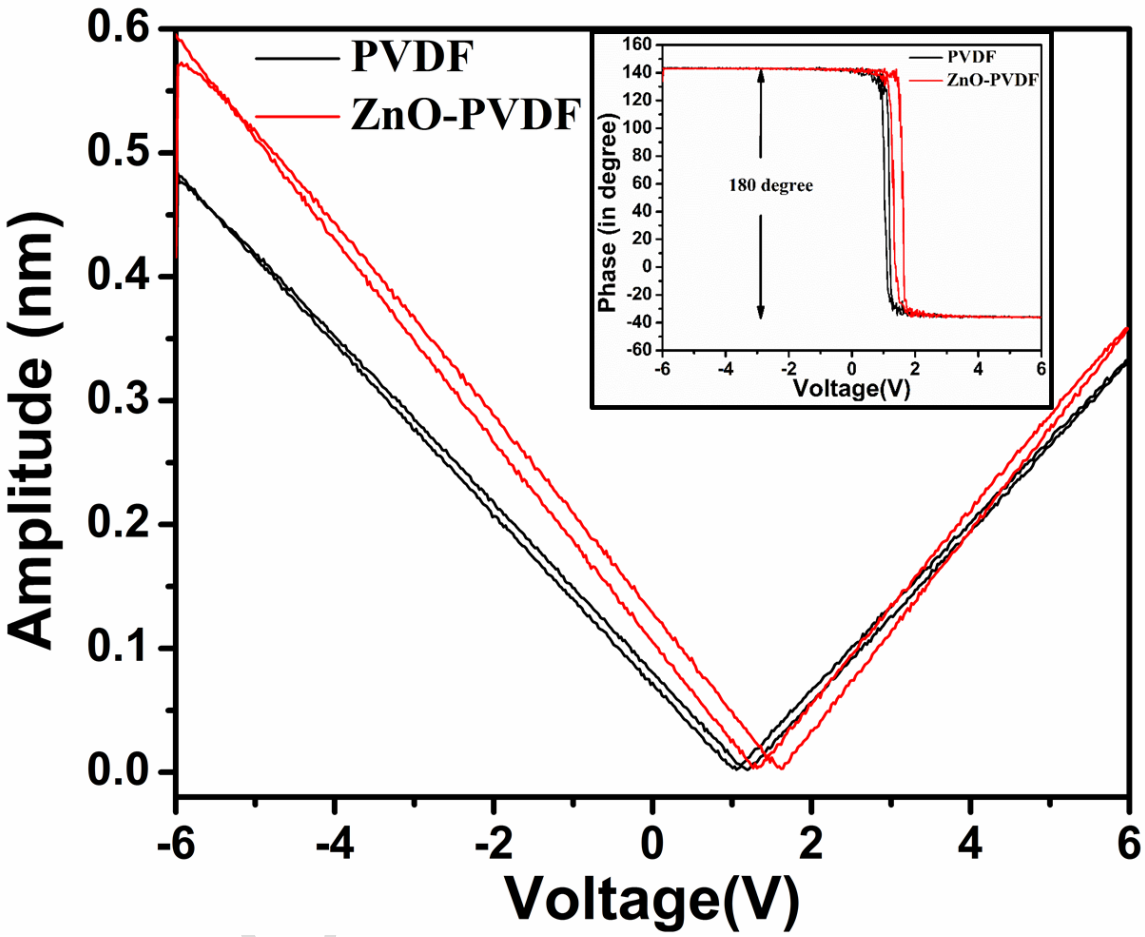


Fig. 3 Piezoelectric displacement of PVDF and ZnO-PVDF nanocomposite film when the voltage from -6 V to +6 V was applied, measured using PFM. The inset of the figure shows the change in PFM phase of the films.

In the PFM measurement, the voltage was applied across the thickness of the film through a conducting tip, and the deformation in the shape of the film was measured. The voltage ranging from -6 V to +6 V was applied across the thickness of the film while the amplitude and phase of the piezoelectric response were recorded. As the voltage was applied

across the thickness of the film, there was a strain produced in the film, and this was picked up by the cantilever of the PFM tip. Again this movement in the cantilever was recorded by the change in the reflected laser light which was incident on the cantilever and can be used to obtain the strain produced in the sample by multiplying the amplitude of the cantilever movement with deflection sensitivity of the cantilever. The calculated amplitude of the deformation in the sample is shown in Fig. 3. It can be seen that classical butterfly loop was obtained for both PVDF and ZnO-PVDF films when the tip biasing voltage was swept from -6 V to +6 V. As the plot is about the amplitude of deformation produced in the sample versus the externally applied voltage, the slope of the curve will give the piezoelectric coefficient d_{33}. It was found that the d_{33} value of ZnO-PVDF as ~81.7 pm/V while that of PVDF ~66.5 pm/V. Further, the inset of Fig. 3 shows the piezoelectric phase hysteresis loops for PVDF and ZnO-PVDF film. A sharp 180° phase change is observed in both cases. This is due to switching of the local electrical domains just below the tip of PFM from antiparallel to parallel or vice versa with respect to the electrical domain of the tip with the application of electrical voltage [45]. It must be noted here that the coercive voltage increases with the incorporation of ZnO into PVDF which signifies a higher ferroelectric property.

Soin et al. [31] showed that PVDF film with higher polarization has more tendency to acquire the charges when two different materials are rubbed together, which leads to higher surface charge density. Therefore, as ZnO-PVDF composite film has more polarizability, it is expected that when we rub ZnO-PVDF film with a material, it will generate more charges as compared to when we rub PVDF with the same material. Apart from the polarizability of the material that affects the triboelectrification, the main important parameter that controls the triboelectrification is the surface roughness. The increase in surface roughness of the films greatly enhances the output of the triboelectric nanogenerators. The enhancement is because of the following reasons: (i) the surface roughness increases the effective surface area of the

contacting surface thereby allowing more charges to reside on the surface leading to more surface charges [10], (ii) when we rub the two surfaces, the rough surface provides more effective friction, and this leads to induce more effective triboelectrification [46], and (iii) the rough surface allows air voids between the two contacting surface, and this increases the effective dielectric constant between the two contacting surfaces [47].

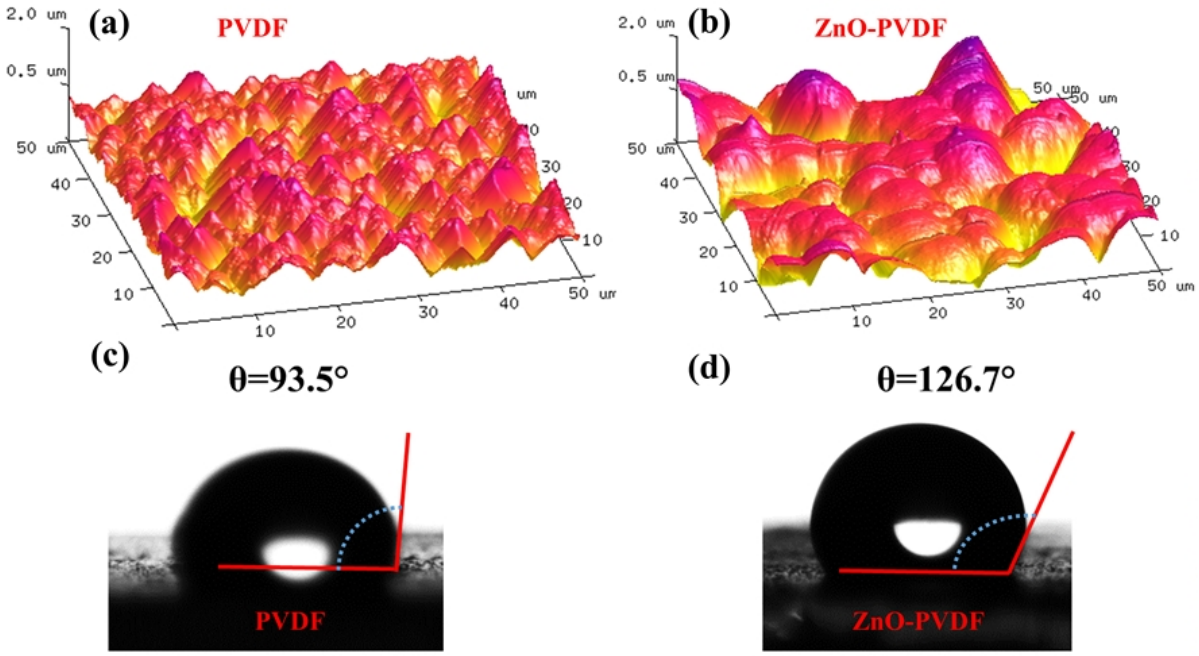


Fig. 4 Atomic force microscopy (AFM) images of (a) PVDF and (b) ZnO-PVDF films. The measured contact angles of water with (c) PVDF and (d) ZnO-PVDF nanocomposite films.

To measure and compare the surface roughness of PVDF and ZnO-PVDF composite films, we did AFM measurement, and the images are shown in Fig. 4(a) and (b). To get the 3-D images of the surfaces and to find the average height deviation from the mean plane (R_a) and root mean square of the average height deviation from the mean plane (R_q) Nanoscope Analysis software was used. The value of R_a was obtained as ~190 nm and ~370 nm; R_q as ~245 nm and ~481 nm for PVDF and ZnO-PVDF, respectively. These results clearly indicate that the ZnO-PVDF nanocomposite film has a more rough surface than PVDF film. Therefore, from the

AFM analysis, it is confirmed that with the addition of ZnO into PVDF, its surface roughness increases which will greatly contribute to enhance the triboelectrification.

The presence of moisture in the atmosphere also hinders the triboelectrification between any two surfaces. The hydrophilic surfaces tend to absorb moisture from the air, and there is the possibility of forming a thin monolayer of water molecule spreading over the surface. Therefore, when the two surfaces rub each other, the presence of water molecules hinder the charge transfer between the two surfaces and leads to a decrease in triboelectrification [48–50]. This hindrance is more pronounced when these triboelectric nanogenerators are used in humid and harsh environments. Therefore, materials having hydrophobic surfaces are preferable for triboelectric nanogenerator applications. The above discussion is about the triboelectrification between any two solid surfaces; however, for solid-liquid interface triboelectrification, it is compulsory to have a hydrophobic surface [51]. Otherwise, the liquid will wet the surface, and the triboelectric nanogenerator will not work properly in consecutive cycles [52–56]. Fig. 4(c) and (d) show the contact angle measurement of pristine PVDF and its nanocomposite film using a water droplet of 2 μL. The measured contact angles for PVDF and ZnO-PVDF nanocomposite films were ~93.5° and ~126.7° respectively. The enhanced contact angle for ZnO-PVDF nanocomposite film indicates that hydrophobic property of PVDF film gets substantially enhanced after ZnO nanorods incorporation in PVDF matrix and therefore, ZnO-PVDF nanocomposite film can be used in the humid environment also.

From the basic definition of triboelectrification it is well known that when two different materials come into contact, the electrons flow from a material having higher surface energy state to materials having lower surface energy state until an equilibrium energy state is achieved [57,58]. The energy states can be quantified with the work function of the materials. Therefore, when the two materials come into contact, the material with lower work function losses electron while the material with the higher work function gains the electron. Thus, the difference in

work function will determine whether there will be more charge transfers or not, and when there is more charge transfer, more power will be generated from the triboelectric nanogenerator. Kelvin probe force microscopy (KPFM) is a powerful tool that can be used to measure the surface potential of the materials, and from the surface potential we can estimate the work function of the material by using the relationship, $\phi_{tip} = \phi_s + \mathbf{eV}_{cpd}$, where ϕ_{tip} and ϕ_s are the work functions of the tip, sample and $\mathbf{V}_{cpd}$ is the contact potential difference between the sample and the tip [59].

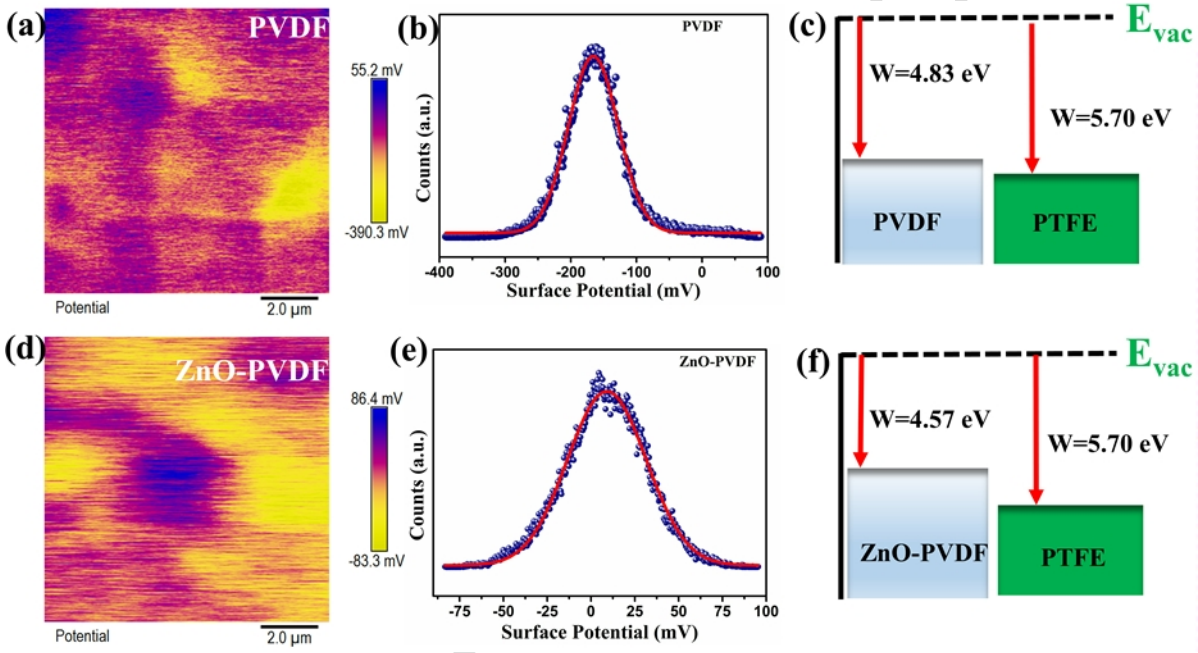


Fig. 5 The surface potential measured by KPFM for (a) pristine PVDF and (b) ZnO-PVDF films. Figure (b) and (e) represents the surface potential distribution curves for PVDF and ZnO-PVDF films while (c) and (f) shows the characteristic energy levels of PVDF and ZnO-PVDF films, calculated from the KPFM measurements with respect to PTFE.

For measuring the surface potentials of the films using the KPFM, the area of the scan was 10 μm × 10 μm which is shown in Fig. 5(a) and (d). The surface potential distribution is plotted, and a Gaussian curve is fitted for both pristine PVDF and ZnO-PVDF nanocomposite film. The two different peaks are obtained (Fig. 5(b) and (e)), one at -165.67 mV for PVDF and the other at 9.03 mV for ZnO-PVDF. The value of ϕ_{tip} as measured using the standard sample was 4.66 eV. Therefore, the work function of pristine PVDF film was estimated as

~4.83 eV while that of ZnO-PVDF as ~4.57 eV. In order to check the triboelectrification, we fabricated PVDF and ZnO-PVDF composite film based triboelectric nanogenerator by coupling with Polytetrafluoroethylene (PTFE). The work function of PTFE is 5.70 eV [60]. Therefore, as the difference in work function of ZnO-PVDF composite film with PTFE is much more than that for PVDF as shown in Fig. 5(c) and (f), more charge transfer will take place for ZnO-PVDF/PTFE to bring to an equilibrium state. Thus, the triboelectric nanogenerator based on ZnO-PVDF/PTFE generated more output power as compared to pristine PVDF/PTFE triboelectric nanogenerator.

The above characterizations point out that addition of ZnO into PVDF will enhance the triboelectrification of PVDF with PTFE because of the following reasons: (i) increase in polarizability, (ii) increased in surface roughness, (iii) increase in hydrophobicity, and (iv) decrease in work function of PVDF. In order to check the enhanced triboelectric effect of PVDF, we fabricated two different triboelectric nanogenerators coupled with PTFE as shown in Fig. 6.

For the device fabrication, one side of each PVDF, ZnO-PVDF and PTFE films were deposited with a layer of aluminum of thickness 100 nm. The electrical contacts were made using copper wires. The mechanical support for sliding was provided by using a wooden structure as shown in Fig. 6(b). By using a double-sided foam tape, the films were attached to the movable wooden structure. The TENG had the effective working area of 2.5 cm × 3 cm. The open circuit voltage measurement of both PVDF/PTFE and ZnO-PVDF/PTFE based TENG is shown in Fig. 7.

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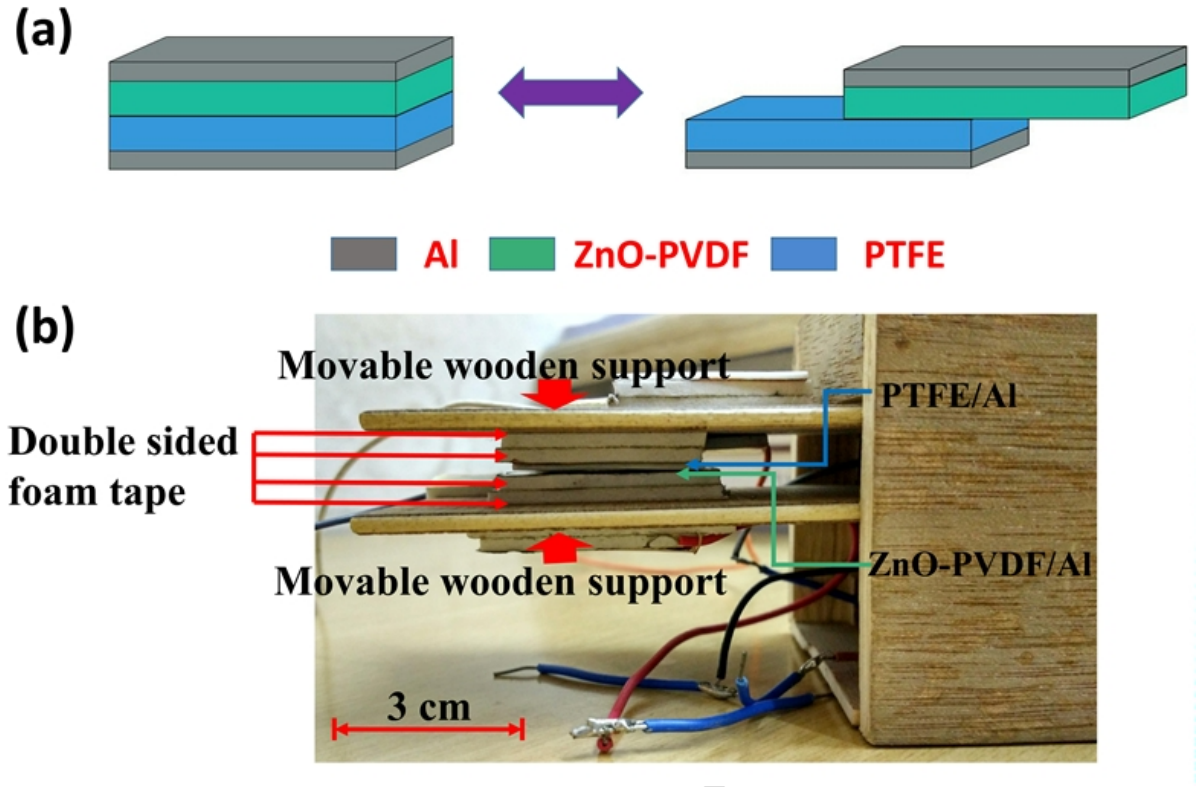


Fig. 6 (a) Schematic of the fabricated triboelectric nanogenerator showing in-plane separation and (b) photographic image of the device showing different components.

The rectified open circuit voltage for PVDF/PTFE based TENG was found to be ~98 V while that of ZnO-PVDF/PTFE based TENG was found to be ~119 V. The measured rectified short circuit current is also shown in Fig. 7(b). The short circuit current from PVDF/PTFE and ZnO-PVDF/PTFE based TENG was obtained as ~1.0 μA and ~1.6 μA. The open circuit voltage and short circuit current measurements clearly indicate that the performance of ZnO-PVDF/PTFE based TENG is more as compared to PVDF/PTFE based TENG.

Further to study the performance, we measured the voltage and current of the TENGs with a variation of the load resistance. It was found that the peak voltage gradually increases as the load resistance increases and become saturated at infinitely higher load resistance. Meanwhile, the current gradually decreased as the resistance increases and become saturated at higher load resistance. The instantaneous output power (P) can be calculated from peak

voltage drop (V), load resistance (R) and effective working area (A) with the relationship, $P = \frac{V^2}{R \times A}$. As it can be seen from Fig. 7(d) and (f), the maximum instantaneous output power of the ZnO-PVDF based TENG was found to be ~10.6 μW/cm^2 while that of PVDF based TENG was found to be ~6.4 μW/cm^2 at 150 MΩ load resistance.

Thus, ZnO-PVDF/PTFE based TENG gives more output power as much as 65.6% enhancement as compared to PVDF/PTFE based TENG. As there was no change in the physical property of PTFE, the enhancement comes only because of the change in properties of PVDF with the incorporation of ZnO.

It is to note that we achieved enhanced performance of triboelectric nanogenerator simply by incorporating ZnO nanorods into the PVDF matrix. No extra efforts were made to modify the surface, nor any complex film preparation technique was used. This improvement with the addition of ZnO is attributed to the increase in β-phase content that leads to the increase in polarization, higher surface roughness and hydrophobicity of PVDF film.

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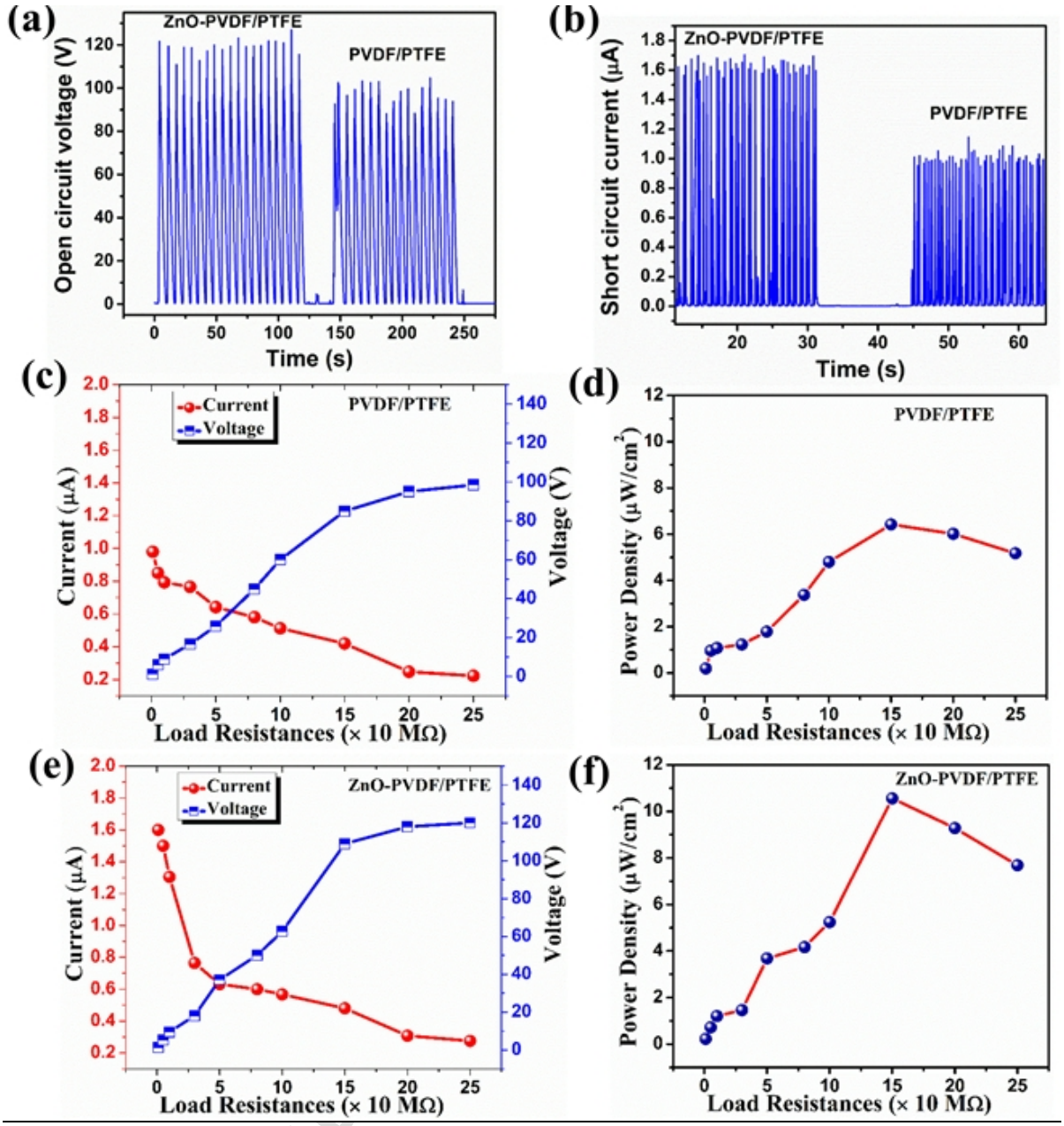


Fig. 7 (a) Open circuit rectified voltage (b) short circuit rectified current for PVDF/PTFE and ZnO-PVDF/PTFE based TENG. Variation of voltage and current with a different load resistance of (c) PVDF/PTFE and (e) ZnO-PVDF/PTFE based TENG. The respective instantaneous output power density with the different load resistance of (d) PVDF/PTFE and (f) ZnO-PVDF/PTFE based TENG.

4. Conclusion

We had synthesized ZnO nanorods, the nanocomposite film of ZnO-PVDF and demonstrated that incorporating ZnO into piezoelectric polymer PVDF is a novel way to enhance the triboelectric effect of PVDF. It is shown that the incorporation of ZnO into PVDF

matrix resulted in the modification of the surface morphology of the film which was confirmed by AFM micrographs, increase in β-phase content and improvement in the hydrophobic nature of the film. It is also confirmed that the work function of PVDF decreases from ~4.83 eV to ~4.57 eV with the incorporation of ZnO into it. These changes lead to the enhancement of the triboelectric effect of ZnO-PVDF/PTFE in comparison to PVDF/PTFE. We have achieved an enhanced output voltage of ~119 V and short circuit current of ~1.6 μA. The instantaneous output power density of ZnO-PVDF/PTFE based TENG was ~10.6 μW/cm^2 which is 65.6% improvement as compared to PVDF/PTFE based TENG. This work provides a simple yet effective approach to enhance the triboelectric effect of a material mainly the ferroelectric polymers.

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References

[1] Wang ZL. Triboelectric nanogenerators as new energy technology for self-powered systems and as active mechanical and chemical sensors. ACS Nano 2013;7:9533–57. doi:10.1021/nn404614z.

[2] Fan FR, Tian ZQ, Lin Wang Z. Flexible triboelectric generator. Nano Energy 2012;1:328–34. doi:10.1016/j.nanoen.2012.01.004.

[3] Park C, Song G, Cho SM, Chung J, Lee Y, Kim EH, et al. Supramolecular-Assembled Nanoporous Film with Switchable Metal Salts for a Triboelectric Nanogenerator. Adv Funct Mater 2017;1701367:1–8. doi:10.1002/adfm.201701367.

[4] Harnchana V, Ngoc H Van, He W, Rasheed A, Park H, Amornkitbamrung V, et al. Enhanced Power Output of a Triboelectric Nanogenerator using Polydimethylsiloxane

Modified with Graphene Oxide and Sodium Dodecyl sulfate. ACS Appl Mater Interfaces 2018;10:25263–72. doi:10.1021/acsami.8b02495.

[5] Liao Q, Zhang Z, Zhang X, Mohr M, Zhang Y, Fecht HJ. Flexible piezoelectric nanogenerators based on a fiber/ZnO nanowires/paper hybrid structure for energy harvesting. Nano Res 2014;7:917–28. doi:10.1007/s12274-014-0453-8.

[6] Ma M, Kang Z, Liao Q, Zhang Q, Gao F, Zhao X, et al. Development, applications, and future directions of triboelectric nanogenerators. Nano Res 2018;11:2951–69. doi:10.1007/s12274-018-1997-9.

[7] Xia X, Chen J, Guo H, Liu G, Wei D, Xi Y, et al. Embedding variable micro-capacitors in polydimethylsiloxane for enhancing output power of triboelectric nanogenerator. Nano Res 2017;10:320–30. doi:10.1007/s12274-016-1294-4.

[8] Lin ZH, Xie Y, Yang Y, Wang S, Zhu G, Wang ZL. Enhanced triboelectric nanogenerators and triboelectric nanosensor using chemically modified TiO2 nanomaterials. ACS Nano 2013;7:4554–60. doi:10.1021/nn401256w.

[9] Li HY, Su L, Kuang SY, Pan CF, Zhu G, Wang ZL. Significant Enhancement of Triboelectric Charge Density by Fluorinated Surface Modification in Nanoscale for Converting Mechanical Energy. Adv Funct Mater 2015;25:5691–7. doi:10.1002/adfm.201502318.

[10] Cheng GG, Jiang SY, Li K, Zhang ZQ, Wang Y, Yuan NY, et al. Effect of argon plasma treatment on the output performance of triboelectric nanogenerator. Appl Surf Sci 2017;412:350–6. doi:10.1016/j.apsusc.2017.03.255.

[11] Shuai X, Zhu P, Zeng W, Hu Y, Liang X, Zhang Y, et al. Highly Sensitive Flexible Pressure Sensor Based on Silver Nanowires-Embedded Polydimethylsiloxane Electrode with Microarray Structure. ACS Appl Mater Interfaces 2017;9:26314–24. doi:10.1021/acsami.7b05753.

[12] Lee S, Lee Y, Kim D, Yang Y, Lin L, Lin ZH, et al. Triboelectric nanogenerator for harvesting pendulum oscillation energy. Nano Energy 2013;2:1113–20. doi:10.1016/j.nanoen.2013.08.007.

[13] Krykpayev B, Farooqui MF, Bilal RM, Vaseem M, Shamim A. A wearable tracking device inkjet-printed on textile. Microelectronics J 2017;65:40–8. doi:10.1016/j.mejo.2017.05.010.

[14] Ha J, Chung J, Kim SM, Kim JH, Shin S, Park JY, et al. Transfer-printable micropatterned fluoropolymer-based triboelectric nanogenerator. Nano Energy 2017;36:126–33. doi:10.1016/j.nanoen.2017.04.009.

[15] Mule AR, Dudem B, Yu JS. High-performance and cost-effective triboelectric nanogenerators by sandpaper-assisted micropatterned polytetrafluoroethylene. Energy 2018;165:677–84. doi:10.1016/J.ENERGY.2018.09.122.

[16] Lee JP, Ye BU, Kim KN, Lee JW, Choi WJ, Baik JM. 3D printed noise-cancelling triboelectric nanogenerator. Nano Energy 2017;38:377–84. doi:10.1016/j.nanoen.2017.05.054.

[17] Zhao P, Soin N, Prashanthi K, Chen J, Dong S, Zhou E, et al. Emulsion Electrospinning

of Polytetrafluoroethylene (PTFE) Nanofibrous Membranes for High-Performance Triboelectric Nanogenerators. ACS Appl Mater Interfaces 2018;10:5880–91. doi:10.1021/acsami.7b18442.

[18] Zhu G, Lin Z-H, Jing Q, Bai P, Pan C, Yang Y, et al. Toward Large-Scale Energy Harvesting by a Nanoparticle-Enhanced Triboelectric Nanogenerator. Nano Lett 2013;13:847–53. doi:10.1021/nl4001053.

[19] Lee JP, Lee JW, Baik JM. The progress of PVDF as a functional material for triboelectric nanogenerators and self-powered sensors. Micromachines 2018;9:30–3. doi:10.3390/mi9100532.

[20] Im JS, Park IK. Mechanically Robust Magnetic Fe3O4 Nanoparticle/Polyvinylidene Fluoride Composite Nanofiber and Its Application in a Triboelectric Nanogenerator. ACS Appl Mater Interfaces 2018;10:25660–5. doi:10.1021/acsami.8b07621.

[21] Xia K, Zhu Z, Zhang H, Xu Z. A triboelectric nanogenerator as self-powered temperature sensor based on PVDF and PTFE. Appl Phys A Mater Sci Process 2018;124:1–7. doi:10.1007/s00339-018-1942-5.

[22] Sahatiya P, Kannan S, Badhulika S. Few layer MoS2 and in situ poled PVDF nanofibers on low cost paper substrate as high performance piezo-triboelectric hybrid nanogenerator: Energy harvesting from handwriting and human touch. Appl Mater Today 2018;13:91–9. doi:10.1016/j.apmt.2018.08.009.

[23] Nunes-Pereira J, Sencadas V, Correia V, Cardoso VF, Han W, Rocha JG, et al. Energy harvesting performance of BaTiO3/poly(vinylidene fluoride-trifluoroethylene) spin coated nanocomposites. Compos Part B Eng 2015;72:130–6. doi:10.1016/j.compositesb.2014.12.001.

[24] Cardoso VF, Minas G, Lanceros-Méndez S. Multilayer spin-coating deposition of poly(vinylidene fluoride) films for controlling thickness and piezoelectric response. Sensors Actuators, A Phys 2013;192:76–80. doi:10.1016/j.sna.2012.12.019.

[25] Zhong H, Xia J, Wang F, Chen H, Wu H, Lin S. Graphene-Piezoelectric Material Heterostructure for Harvesting Energy from Water Flow. Adv Funct Mater 2017;27:1604226. doi:10.1002/adfm.201604226.

[26] Lee J-H, Yoon H-J, Kim TY, Gupta MK, Lee JH, Seung W, et al. Micropatterned P(VDF-TrFE) Film-Based Piezoelectric Nanogenerators for Highly Sensitive Self-Powered Pressure Sensors. Adv Funct Mater 2015;25:3203–9. doi:10.1002/adfm.201500856.

[27] Martins P, Lopes AC, Lanceros-Mendez S. Electroactive phases of poly(vinylidene fluoride): Determination, processing and applications. Prog Polym Sci 2014;39:683–706. doi:10.1016/j.progpolymsci.2013.07.006.

[28] Lovinger AJ. Ferroelectric Polymers. Science (80-) 1983;220:1115–21. doi:10.1126/science.220.4602.1115.

[29] Salimi A, Yousefi AA. Analysis Method: FTIR studies of β-phase crystal formation in stretched PVDF films. Polym Test 2003;22:699–704. doi:10.1016/S0142-9418(03)00003-5.

[30] Martins P, Lopes AC, Lanceros-Mendez S. Electroactive phases of poly(vinylidene fluoride): Determination, processing and applications. Prog Polym Sci 2014;39:683–706. doi:10.1016/j.progpolymsci.2013.07.006.

[31] Soin N, Zhao P, Prashanthi K, Chen J, Ding P, Zhou E, et al. High performance triboelectric nanogenerators based on phase-inversion piezoelectric membranes of poly(vinylidene fluoride)-zinc stannate (PVDF-ZnSnO3) and polyamide-6 (PA6). Nano Energy 2016;30:470–80. doi:10.1016/j.nanoen.2016.10.040.

[32] Chen F, Wu Y, Ding Z, Xia X, Li S, Zheng H, et al. A novel triboelectric nanogenerator based on electrospun polyvinylidene fluoride nanofibers for effective acoustic energy harvesting and self-powered multifunctional sensing. Nano Energy 2019;56:241–51. doi:10.1016/J.NANOEN.2018.11.041.

[33] Willander M, Nur O, Zhao QX, Yang LL, Lorenz M, Cao BQ, et al. Zinc oxide nanorod based photonic devices: recent progress in growth, light emitting diodes and lasers. Nanotechnology 2009;20:332001. doi:10.1088/0957-4484/20/33/332001.

[34] Wang ZL. ZnO nanowire and nanobelt platform for nanotechnology. Mater Sci Eng R Reports 2009;64:33–71. doi:10.1016/j.mser.2009.02.001.

[35] Lee SG, Ha J-W, Sohn E-H, Park IJ, Lee S-B. Enhancement of polar crystalline phase formation in transparent PVDF-CaF 2 composite films. Appl Surf Sci 2016;390:339–45. doi:10.1016/j.apsusc.2016.08.090.

[36] Gomes J, Serrado Nunes J, Sencadas V, Lanceros-Mendez S. Influence of the β-phase content and degree of crystallinity on the piezo- and ferroelectric properties of poly(vinylidene fluoride). Smart Mater Struct 2010;19:065010. doi:10.1088/0964-1726/19/6/065010.

[37] Chen X, Li X, Shao J, An N, Tian H, Wang C, et al. High-Performance Piezoelectric Nanogenerators with Imprinted P(VDF-TrFE)/BaTiO $_3$ Nanocomposite Micropillars for Self-Powered Flexible Sensors. Small 2017;13:1604245. doi:10.1002/smll.201604245.

[38] Singh HH, Singh S, Khare N. Design of flexible PVDF/NaNbO3/RGO nanogenerator and understanding the role of nanofillers in the output voltage signal. Compos Sci Technol 2017;149:127–33. doi:10.1016/j.compscitech.2017.06.013.

[39] Singh HH, Singh S, Khare N. Enhanced β-phase in PVDF polymer nanocomposite and its application for nanogenerator. Polym Adv Technol 2018;29:143–50. doi:10.1002/pat.4096.

[40] Schmidt VH, Polasik J, Lediaev L, Hallenberg J. Actuators based on PVDF sheets with flexible PEDOT polymer electrodes. 2005 12th Int Symp Electrets 2005:378–81. doi:10.1109/ISE.2005.1612402.

[41] Zhang X, Huang X, Li C, Jiang H. Dye-sensitized solar cell with energy storage function through PVDF/ZnO nanocomposite counter electrode. Adv Mater 2013;25:4093–6. doi:10.1002/adma.201301088.

[42] Lee SG, Ha J, Sohn E, Park IJ, Lee S. Enhancement of polar crystalline phase formation in transparent PVDF-CaF 2 composite films. Appl Surf Sci 2016;390:339–45. doi:10.1016/j.apsusc.2016.08.090.

[43] Kang HB, Han CS, Pyun JC, Ryu WH, Kang CY, Cho YS. (Na,K)NbO3 nanoparticle-embedded piezoelectric nanofiber composites for flexible nanogenerators. Compos Sci Technol 2015;111:1–8. doi:10.1016/j.compscitech.2015.02.015.

[44] Ahmad Z. Polymer Dielectric Materials. Dielectr. Mater., InTech; 2012. doi:10.5772/50638.

[45] Maji S, Sarkar PK, Aggarwal L, Ghosh SK, Mandal D, Sheet G, et al. Self-oriented β-crystalline phase in the polyvinylidene fluoride ferroelectric and piezo-sensitive ultrathin Langmuir–Schaefer film. Phys Chem Chem Phys 2015;17:8159–65. doi:10.1039/C5CP00218D.

[46] Mallineni SSK, Behlow H, Dong Y, Bhattacharya S, Rao AM, Podila R. Facile and robust triboelectric nanogenerators assembled using off-the-shelf materials. Nano Energy 2017;35:263–70. doi:10.1016/j.nanoen.2017.03.043.

[47] Fan FR, Lin L, Zhu G, Wu W, Zhang R, Wang ZL. Transparent triboelectric nanogenerators and self-powered pressure sensors based on micropatterned plastic films. Nano Lett 2012;12:3109–14. doi:10.1021/nl300988z.

[48] Li Z, Shen J, Abdalla I, Yu J, Ding B. Nanofibrous membrane constructed wearable triboelectric nanogenerator for high performance biomechanical energy harvesting. Nano Energy 2017;36:341–8. doi:10.1016/j.nanoen.2017.04.035.

[49] Nguyen V, Zhu R, Yang R. Environmental effects on nanogenerators. Nano Energy 2015;14:49–61. doi:10.1016/j.nanoen.2014.11.049.

[50] Cui C, Wang X, Yi Z, Yang B, Wang X, Chen X, et al. Flexible Single-Electrode Triboelectric Nanogenerator and Body Moving Sensor Based on Porous Na2CO3/Polydimethylsiloxane Film. ACS Appl Mater Interfaces 2018;10:3652–9. doi:10.1021/acsami.7b17585.

[51] Pan L, Wang J, Wang P, Gao R, Wang Y, Zhang X, et al. Liquid-FEP-based U-tube triboelectric nanogenerator for harvesting water-wave energy 2018;11:4062–73.

[52] Xu W, Huang LB, Wong MC, Chen L, Bai G, Hao J. Environmentally Friendly Hydrogel-Based Triboelectric Nanogenerators for Versatile Energy Harvesting and Self-Powered Sensors. Adv Energy Mater 2017;7:1–8. doi:10.1002/aenm.201601529.

[53] Kil Yun B, Soo Kim H, Joon Ko Y, Murillo G, Hoon Jung J. Interdigital electrode based triboelectric nanogenerator for effective energy harvesting from water. Nano Energy 2017;36:233–40. doi:10.1016/j.nanoen.2017.04.048.

[54] Lin M-F, Parida K, Cheng X, Lee PS. Flexible Superamphiphobic Film for Water Energy Harvesting. Adv Mater Technol 2016:1600186. doi:10.1002/admt.201600186.

[55] Nahian SA, Cheedarala RK, Ahn KK. A study of sustainable green current generated by the fluid-based triboelectric nanogenerator (FluTENG) with a comparison of contact and sliding mode. Nano Energy 2017;38:447–56. doi:10.1016/j.nanoen.2017.06.012.

[56] Xiong J, Lin M-F, Wang J, Gaw SL, Parida K, Lee PS. Wearable All-Fabric-Based Triboelectric Generator for Water Energy Harvesting. Adv Energy Mater 2017;1701243:1701243. doi:10.1002/aenm.201701243.

[57] Bai P, Zhu G, Zhou YS, Wang S, Ma J, Zhang G, et al. Dipole-moment-induced effect on contact electrification for triboelectric nanogenerators. Nano Res 2014;7:990–7. doi:10.1007/s12274-014-0461-8.

[58] Kim HJ, Yim EC, Kim JH, Kim SJ, Park JY, Oh IK. Bacterial Nano-Cellulose Triboelectric Nanogenerator. Nano Energy 2017;33:130–7. doi:10.1016/j.nanoen.2017.01.035.

[59] Lee JW, Cho HJ, Chun J, Kim KN, Kim S, Ahn CW, et al. Robust nanogenerators based on graft copolymers via control of dielectrics for remarkable output power enhancement. Sci Adv 2017;3:1–10. doi:10.1126/sciadv.1602902.

[60] Trigwell S, Captain JG, Arens EE, Quinn JW, Calle CI. The use of tribocharging in the electrostatic beneficiation of lunar simulant. IEEE Trans Ind Appl 2009;45:1060–7. doi:10.1109/TIA.2009.2018976.

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Highlights

- Achieved 65.6% more instantaneous output power in nanocomposite film as compared to bare film based TENG.
- Surface roughness, polarizability and hydrophobicity plays an important role in the performance of TENG.
- Simple yet effective way of enhancing the triboelectric property of ferroelectric film (PVDF).
- The localised ferroelectric property of PVDF was studied using PFM.

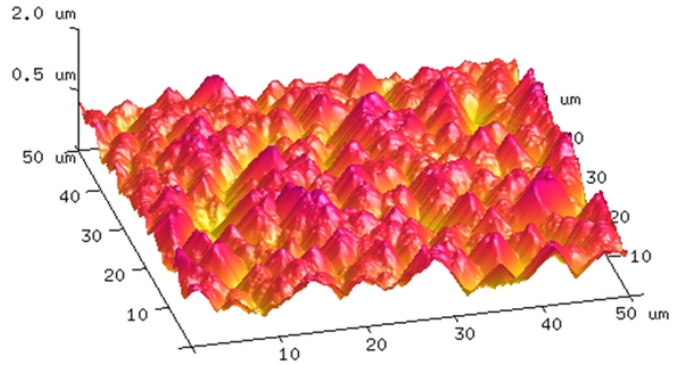
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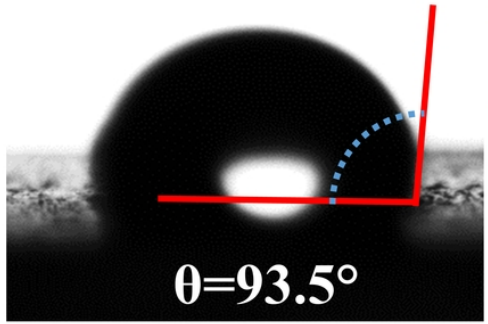
Surface Roughness

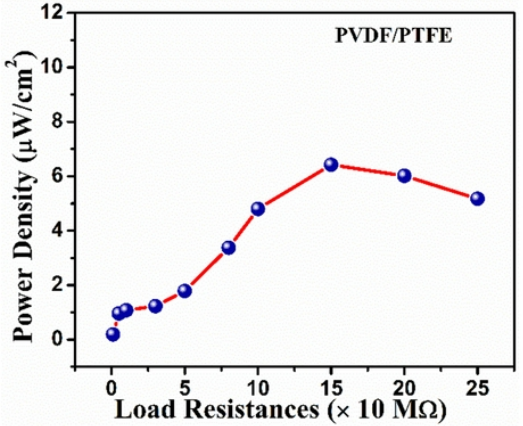
Contact Angle

Output Power Density

PVDF







ZnO-PVDF

